Secret Communication with Plausible Deniability

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Who Ate the Cookies?

Two possibilities



The jar is empty!



the dog?

or



Brother?

"You're grounded – I saw the crumbs!"

psst... (private)



Sister
(saw everything)



Mom

Everyone sees: crumbs on his shirt
(but he also had cookies at lunch...)

Brother blames the crumbs – not his sister.

The trick: the secret is safe because the *public* evidence can explain the punishment.

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Why Should Economists Care?

In the kitchen	In the real world	In our paper
Mom	regulator, police, army command	**decision maker**
Sister	whistleblower, informant, spy	**source**
crumbs on the shirt	audit report, public complaint	**public evidence**
Brother	the company, the gang, the enemy	**whoever might retaliate**

The source needs **protection**: getting caught can mean a lost job, prison, or worse.

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The Big Question

- Mom wants Sister's information. Sister wants to stay safe.
- Two kinds of **protection** for Sister:
  - **Secrecy**: if Brother overhears the whisper, it tells him nothing.
  - **Plausible deniability**: whatever Mom does, the public evidence can explain it.
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Part 1: The Cookie Model

Setting the Scene

- **What happened** (only Sister knows):
  the dog or Brother 50–50
- **What everyone sees**:
  crumbs on his shirt or a clean shirt
- **What Mom does**:
  let it go $<$ scold $<$ ground

	clean shirt	crumbs
dog	80%	20%
Brother	20%	80%

Even if the dog did it, Brother may have crumbs from his lunch (20%).

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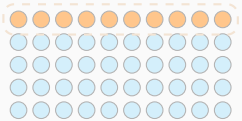
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Crumbs Are a Clue, Not Proof

Imagine 100 kitchens just like ours:

50 kitchens: the dog did it

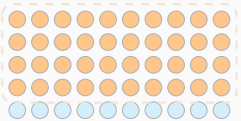


10 with crumbs

crumbs

clean shirt

50 kitchens: Brother did it

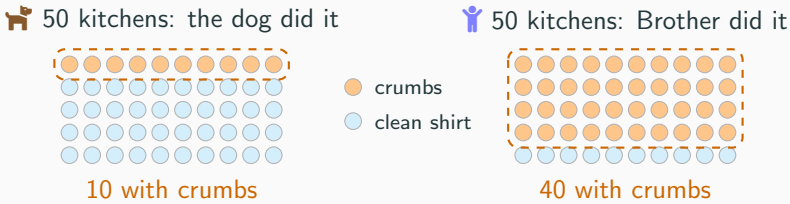


40 with crumbs

- Crumbs show up in $10 + 40 = 50$ kitchens. Brother did it in 40 of them: **80%**.
- A clean shirt shows up in $40 + 10 = 50$ kitchens. Brother did it in 10: **20%**.

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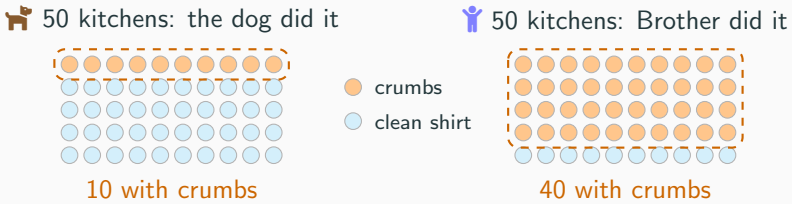
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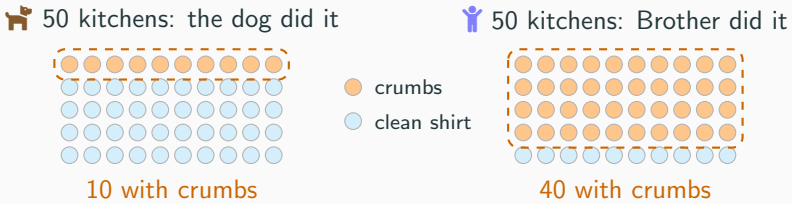
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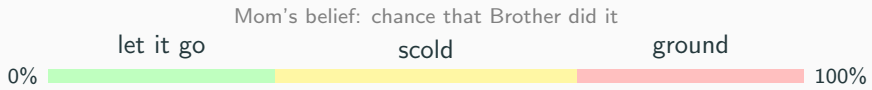
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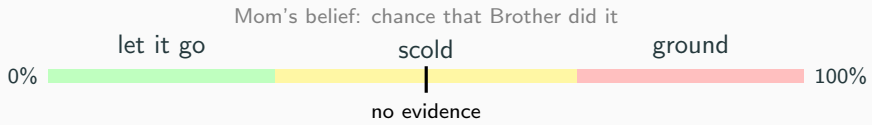
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- What everyone *does* know: at 50–50 she scolds, and she gets harsher the guiltier Brother looks.

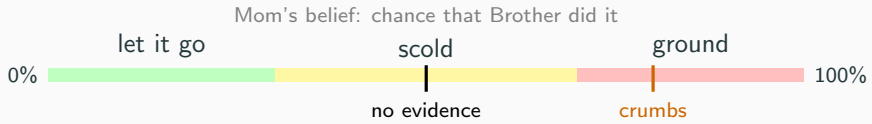
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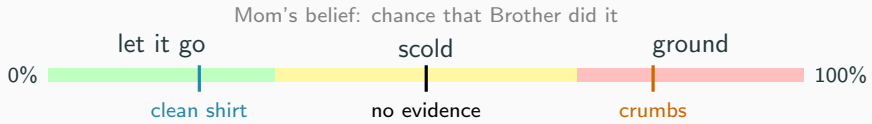
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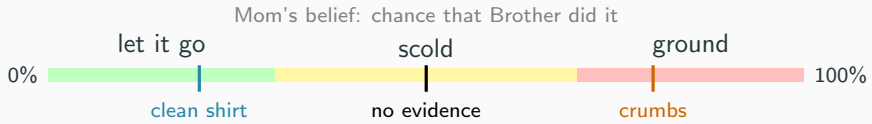
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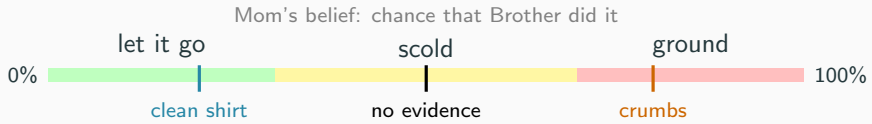
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What Can the Shirt Explain?

	let it go	scold	ground
crumbs	✗	✓	✓
clean shirt	✓	✓	✗

- Crumbs make Brother look guiltier than 50–50. A soft mom might scold, a strict mom might ground – but *no* mom lets him off.
- A clean shirt works the other way: no mom grounds him.
- So if Mom lets a crumb-covered Brother off, there must be a **hidden reason**...

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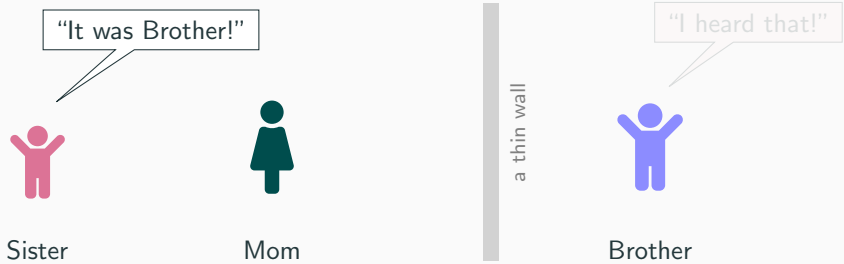
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Part 2: Sister's Whisper

Plan A: Just Tell Mom



- Mom learns everything – but the whisper *itself* is evidence. Overheard once, Sister is caught.
- Fix: make the whisper **meaningless on its own**. This is **secrecy**.

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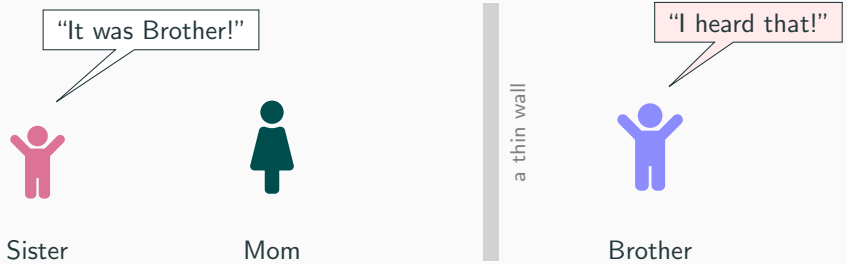
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Secrecy: Speak in Code

- Mom and Sister share a secret **code book** 📖. Sister only says animal names: **Toucan**, **Falcon**, **Manatee**, ...
- **Secrecy rule:** each word is said *equally often*, whether the dog or Brother did it.
- So a word alone is as useless as a coin flip: overhearing "**Toucan**" tells Brother *nothing*.
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Plan B: The Perfect Code?

Code book **Toucan** = "the shirt is telling the truth"
Falcon = "the shirt is lying"

	clean + **Toucan**	crumbs + **Falcon**	clean + **Falcon**	crumbs + **Toucan**
dog	80%	20%	0	0
Brother	0	0	20%	80%

- **Secret?** **Toucan** is said 80% of the time in *both* worlds, **Falcon** 20% in both. ✓
- **Useful?** Each combination points to one world: Mom always learns the truth!
- Too good to be true. . . ?

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Plan B Gets Sister Caught

Case 1: crumbs + **Falcon**

Mom knows: the dog did it ⇒ lets Brother go.

Everyone: *"Crumbs all over him, and no punishment? Someone vouched for him. . . "*

Case 2: clean shirt + **Falcon**

Mom knows: Brother did it ⇒ grounds him.

Brother: *"My shirt is spotless! Who told on me?!"*

Secret words, revealing actions: Mom acts *against* the shirt.

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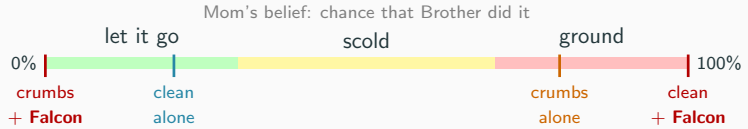
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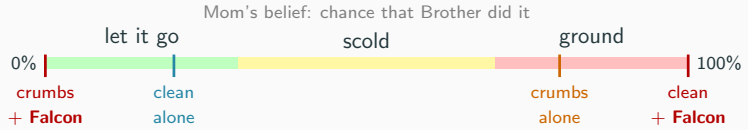
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Plausible Deniability

The second protection

Whatever Mom does must be explainable by **the shirt alone**.

- After **crumbs**: the whisper may change Mom's mind, but Brother must still look *at least* 50–50 guilty.
- After a **clean shirt**: Brother must still look *at most* 50–50 guilty.

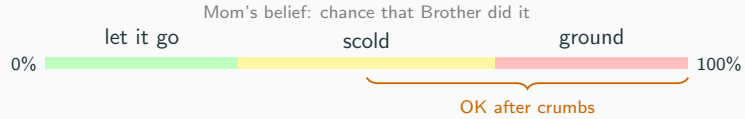
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Plausible Deniability

The second protection

Whatever Mom does must be explainable by **the shirt alone**.

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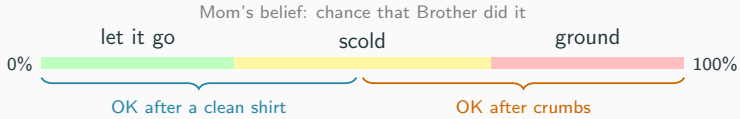
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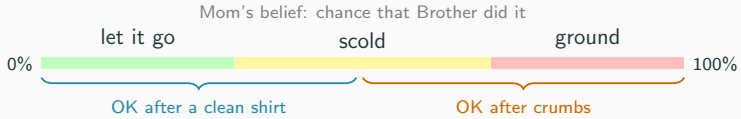
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Plan C: Toucan and Manatee

New code book **Toucan** = "the shirt is right"
Manatee = "nothing – ignore me"

	clean + **Toucan**	crumbs + **Manatee**	clean + **Manatee**	crumbs + **Toucan**
dog	60%	20%	20%	0
Brother	0	20%	20%	60%

- **Secret?** **Toucan** is said 60% of the time in *both* worlds, **Manatee** 40% in both. ✓
- When the shirt is misleading, Sister never flags it. She just says **Manatee**.

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Plan C Keeps Sister Safe

Mom sees + hears	Mom believes	Mom does	Shirt explains it?
crumbs + **Toucan**	Brother, for sure	ground	✓
crumbs + **Manatee**	50–50	scold	✓
clean + **Manatee**	50–50	scold	✓
clean + **Toucan**	the dog, for sure	let it go	✓

Mom never acts *against* the shirt ⇒ nobody can tell a whisper happened.

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Why Do We Need the Manatee?

- Sometimes the shirt is **misleading**: crumbs when the dog did it (20%), a clean shirt when Brother did it (20%).
- Sister can't flag these – that was exactly Plan B's mistake.
- To hide a misleading shirt, she mixes it with an *equal* number of honest ones: crumbs & the dog (20%) + crumbs & Brother (20%) $\Rightarrow$ 50–50.
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Plan	Secret?	Deniable?	Mom is sure...
No whisper	✔	✔	never
A: "It was Brother!"	✘	✘	always
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C: **Toucan** / **Manatee**	✔	✔	60% of the time

Our result: no safe code beats Plan C – for *every* mom, strict or soft.

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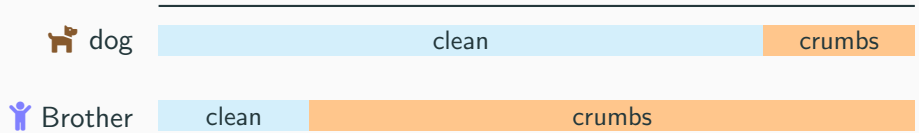
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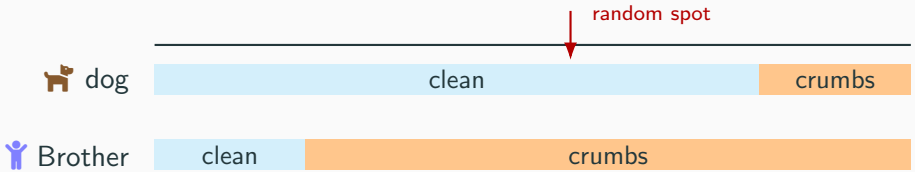
Part 3: Where Does the Code Come From?

The Paint-Strip Trick



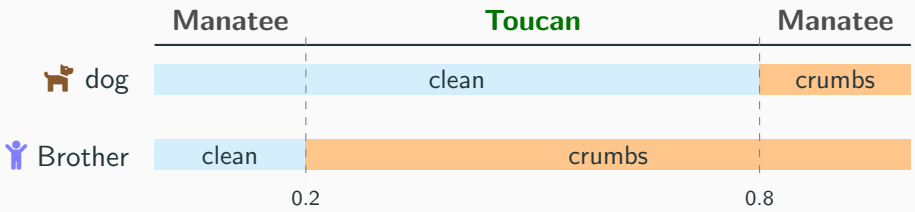
- Each world is a strip; the share of each color is the chance of that shirt.
- Think of the shirt as the color at a **random spot**, which Sister secretly sees.
- Sister whispers only the **zone**. The spot is equally likely to land anywhere in *both* worlds $\Rightarrow$ **secrecy for free**.
- Middle zone: the colors differ, so the shirt now tells all. Outer zones: same color in both rows, so nothing new.

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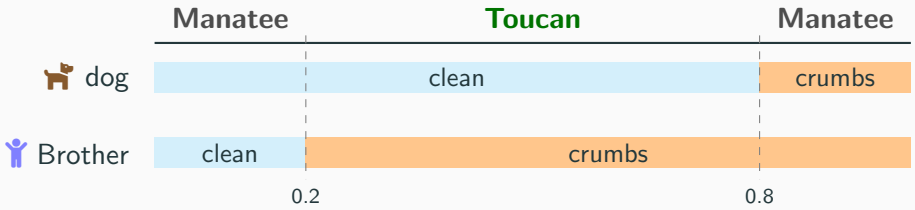
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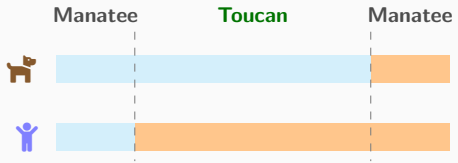
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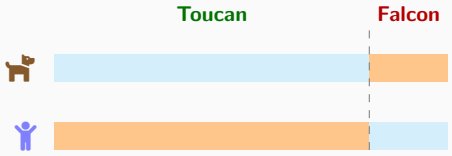
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Order Matters

Plan C: same order



Plan B: Brother's strip flipped

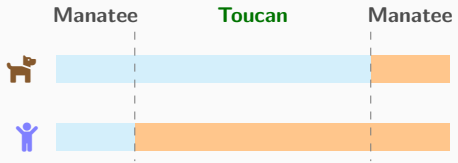


clean shirt crumbs

Rule of thumb: paint every strip in the *same order* – clean first, crumbs last.

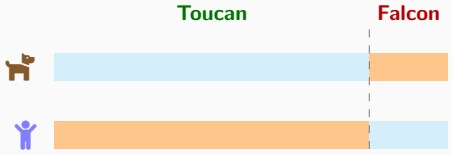
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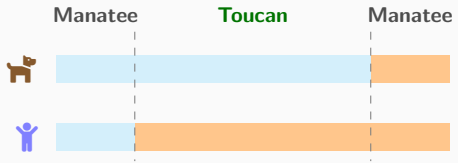
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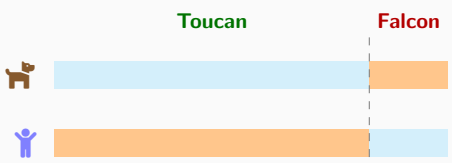
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Part 4: Beyond the Kitchen

The Real Research

- Today we kept things as small as possible:
  - **2 possibilities**: the dog or Brother
  - **2 kinds of evidence**: a clean shirt or crumbs
- In the paper, we answer the same question with **more** of both, as long as they are **ordered**:
  - possibilities: Brother ate *none* (all the dog) < Brother ate *half* < Brother ate *all*
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- The lesson carries over: the best safe code still **lines up the strips** in the same order.

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Thank You!

Questions?

Joint work with Xiaoyu Cheng (FSU) and Pak Hang (Michael) Tam (UNC)